

Question 7 (CNNs, 6 points + 2 extra credit points)

- (a) **(1 point)** Give a reason why one would use a 1×1 convolution. Hint: what does performing a 1×1 convolution achieve in terms of the resulting output volume? (1 sentence)

Solution: A (1×1) convolution would be useful if you wanted to reduce the number of channels while preserving the spatial dimensions.

- (b) **(1 point)** What would you set the padding of a 2D CONV layer to be (as a function of the filter width f) to ensure that the output has the same dimension as the input? Assume the stride is 1. (1 formula)

Solution: $\frac{f-1}{2}$

- (c) **(2 points)** You have an input volume of $32 \times 32 \times 3$. What are the dimensions of the resulting volume after convolving a 5×5 kernel with zero padding, stride of 1, and 2 filters? (1 formula)

Solution: $(28 \times 28 \times 2)$

- (d) **(2 point)** How many weights and biases would the layer defined in (c) have? (1 formula)

Solution: Number of parameters = $(F * F * D) * K + K = 5 * 5 * 3 * 2 + 2 = 152$

- (e) **(Extra credit: 1 point)** You want to process time-series data with a 1D CONV that has the same configuration as the layer presented in (c) but with a kernel of size 5. The input volume of shape $T \times 3$ models three fluctuating values over time. How many weights and biases does this layer have? Assume the same configuration (padding, stride, number of filters) as in (c) and show your work.

Solution: Number of parameters = $(F * D) * K + K = (5 * 3) * 2 + 2 = 32$

- (f) **(Extra credit: 1 point)** You want to process a video with a 3D CONV that has the same configuration as the layer presented in (c) but with kernel of shape $5 \times 5 \times 5$. The input video can be seen as a sequence of images indexed by time, i.e. a volume of shape $W \times H \times T \times 3$. How many weights and biases does this layer have? Assume the same configuration (padding, stride, number of filters) as in (c) and show your work.

Solution: Number of parameters = $(F * F * F * D) * K + K = 5 * 5 * 5 * 3 * 2 + 2 = 752$